Inverse-image derivability transfers through the box

The Open Logic Project contributors

openlogic:normal-modal-logic/completeness:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-normal-modal-logic-completeness/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the pool's line pair for this lemma was twelve rows early (drift); the span is anchored to the label line 1214 at generation.

Editorial note. The label lem:box2 names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Box1: boxing preserves derivability from premise sets. Same section, \ol-label{lem:box1} (source line 1155). Line 1219: the hypothesis re-packed by box1.
- Monotonicity of derivability in the premise set. Proof theory part: weakening for Sigma-derivability. Line 1221: shrinking the boxed premises from Box Box^{-1} Gamma to Gamma.

Source reference. Original lemma lem-box1: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 1199–1199. Prerequisite pointer invoked at proof line(s) 1221; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *If* $\Box^{-1}\Gamma \vdash [\Sigma]A$ *then* $\Gamma \vdash [\Sigma]\Box A$.

Proof. Suppose $\Box^{-1}\Gamma \vdash [\Sigma]A$; then by lem-box1, $\Box\Box^{-1}\Gamma \vdash \Box A$. But since $\Box\Box^{-1}\Gamma \subseteq \Gamma$, also $\Gamma \vdash [\Sigma]\Box A$ by monotonicity. □